

CAD/CAM

BRANCH: MECHANICAL ENGG. (ME)

COURSE CODE: ME401

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for CAD/CAM (ME401) under the Mechanical Engg. (ME) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Computer Graphics & Modeling

Introduction to CAD, hardware, raster graphics, transformations (translation, scaling, rotation). Geometric modeling: Wireframe, surface, and solid modeling. Bezier, B-Spline curves.

Unit 2: Numerical Control (NC) & CNC

NC machine components, MCU, coordinate systems. CNC machining centers, G-code and M-code programming. Manual and computer-assisted part programming.

Unit 3: Group Technology & FMS

Part families, classification and coding (Opitz system). Flexible Manufacturing Systems (FMS): layouts, material handling, AGVs.

Unit 4: Rapid Prototyping & CAPP

Introduction to Additive Manufacturing (SLA, FDM, SLS). Computer-Aided Process Planning (CAPP): Retrieval and Generative approaches.

2D Rotation Matrix: $\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$. G00: Rapid linear positioning; G01: Linear interpolation.

Key Formulas & Exam Pointers



